APPENDIX

Appendix A — Mathematical Notation

This section defines the notation used throughout the Marketing Mix Model.

Symbol	Definition
t	time index
g	geographic unit (region)
b	brand or product
i	marketing channel
j	control variable
$Y_{t,g,b}$	observed KPI
$X_{i,t,g,b}$	marketing exposure
$Z_{j,t,g,b}$	control variable
$S_{i,t,g,b}$	marketing spend
$Adstock_{i,t}$	carryover-adjusted exposure
β_i	channel effectiveness coefficient
γ_i	saturation parameter
θ_i	adstock retention parameter
σ	observation noise

Appendix B — Complete Model Specification

The complete MMM model is defined as:

$$Y_{t,g,b} \sim Normal(\mu_{t,g,b}, \sigma)$$

Where

$$\mu_{t,g,b} = Base_{t,g,b} + Media_{t,g,b} + Control_{t,g,b}$$

Baseline Component

$$Base_t = \alpha + \beta_{trend} t + Seasonality_t$$

Seasonality

$$Season_t = \sum_{k=1}^{K} \left(a_k sin\left(\frac{2\pi kt}{52}\right) + b_k cos\left(\frac{2\pi kt}{52}\right) \right)$$

Adstock Transformation

$$Adstock_{i,t} = X_{i,t} + \theta_i Adstock_{i,t-1}$$

Saturation Function

$$S(X) = \frac{X}{\gamma_i + X}$$

Media Contribution

$$Media_t = \sum_{i} \beta_i S(Adstock_{i,t})$$

Appendix C — Parameter Priors

The Bayesian model uses the following priors.

Parameter	Prior
Intercept	Normal(0,1)

Parameter	Prior
Trend coefficient	Normal(0,0.1)
Seasonality coefficients	Normal(0,0.1)
Media coefficients	HalfNormal(1)
Control coefficients	Normal(0,0.5)
Adstock parameter	Beta(2,2)
Saturation parameter	HalfNormal
Noise variance	HalfNormal

These priors ensure reasonable parameter ranges and improve estimation stability.

Appendix D — Tensor Structure

All modelling inputs are represented as tensors.

Primary tensor dimensions:

(T, G, SB, C)

Where

Dimension	Meaning
T	time periods
G	geographic regions
SB	sub-brands
C	marketing channels

Example Tensor Shapes

Tensor	Shape
Y	(T, G, SB)
X_media	(T, G, SB, C)

Tensor	Shape
X_control	(T, G, SB, K)
X_spend	(T, G, SB, C)

Appendix E — ROI Calculation Example

Assume the following example results.

Channel	Spend	Contribution Volume
Email	10,000	2,000
Calls	50,000	5,000

Assume revenue per unit:

$20

Revenue Contribution

$$Revenue = Contribution_{volume} \times RevenuePerUnit$$

Example:

Channel	Revenue Contribution
Email	40,000
Calls	100,000

ROI Calculation

$$ROI = \frac{Revenue}{Spend}$$

Channel	ROI
Email	4.0
Calls	2.0

Appendix F — ROI Stability Classification

ROI uncertainty is measured using posterior intervals.

$$StabilityRatio = \frac{ROI_{p95} - ROI_{p05}}{|\ ROI_{mean}\ |}$$

Classification rules:

Stability Ratio	Classification
< 1	stable
1–2	moderate

2 | unstable |

This helps identify channels with uncertain estimates.

Appendix G — Example Output Files

The modelling pipeline produces the following outputs.

File	Description
roi.csv	channel ROI summary
media_contribution.csv	contribution time series
base_contribution.csv	baseline KPI
marginal_roi_curves.csv	response curves
roi_key_points.csv	key investment points
fit_metrics.csv	model diagnostics

Appendix H — Example ROI Output

Example structure of roi.csv.

media	spend	contribution	revenue	ROI
clm_call	7,581,530	464,467	9,566,388	1.26

media	spend	contribution	revenue	ROI
email	11,469	3,158	65,045	5.67
mass_email	239,066	34,747	715,673	2.99

Appendix I — Glossary of Terms

Term	Definition
Adstock	carryover effect of marketing
Saturation	diminishing returns response
Contribution	incremental KPI attributed to a channel
ROI	revenue return per unit of spend
Marginal ROI	incremental return from additional spend
Baseline	demand independent of marketing

Appendix J — Model Limitations

All statistical models have limitations.

Key considerations include:

• model results depend on data quality
• insufficient variation can weaken causal inference
• correlated marketing channels may introduce uncertainty
• results should be interpreted alongside domain knowledge

Appendix K — Reproducibility

Each model run is reproducible using:

data snapshot
info configuration
model version
random seed

These elements ensure full auditability of modelling results.